

The Name-Leash Hypothesis: Maternal Separation and the Origins of Displaced Reference

Sergey Maslov

Independent researcher

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Abstract

How did human communication first escape the "here and now"? The capacity for displaced reference — referring to entities not present in the immediate perceptual environment — is widely recognized as a defining feature of human language (Hockett, 1960), and Bickerton (2009) rightly placed it at the center of the language origins problem. Yet the dominant scenario, which locates displacement in the need to recruit coalition members to distant food sources, does not compellingly explain why an offline representational system would be selected when pointing, leading, and pantomime would suffice.

The Name-Leash hypothesis proposes that the first displaced signal was not a food reference but a **proper name** — an individualized vocalization addressed to an absent infant. In cooperative breeding contexts characteristic of early *Homo* (Hrdy, 2009; Burkart et al., 2025), mothers regularly transferred infants to allomothers and departed to forage, generating an irreducible affective pressure — separation anxiety directed at a specific, non-fungible child — that no perceptually grounded gesture could resolve.

The hypothesis identifies a semiotic mechanism for this transition: when the referent of an indexical signal disappears, the index "breaks," creating pressure for a vocal substitute that maintains the referent's mental representation in absence. The proper name fills this role as a **repaired index** — preserving the singularity of indexical pointing while acquiring the context-independence of a symbol. In Kripke's (1980) terms, the name functions as a rigid designator: it carries no descriptive content requiring environmental verification, making it the only signal type that operates precisely when there is nothing to perceive.

The mechanism is grounded in the physiological platform created by habitual upright walking (decoupled respiration, enhanced vagal regulation), in the neural architecture of invariant identity representation (concept cells; dedicated name-retrieval systems in the left temporal pole), and in the ethology of cooperative breeding. Developmental evidence — the coincidence of peak solitary babbling with separation anxiety, Freud's Fort/Da, and Porshnev's analysis of the first word as self-regulatory — independently supports the proposed dynamic. Comparative data show that the two prerequisites for the mechanism — vocal individuation and vocal mimesis — exist separately in other species but converge only in cooperative-breeding hominins.

The proto-name, born as a tool of affective self-regulation, already carries relational (genitive) structure and functions as an implicit promise of return — suggesting that reference and social obligation may have co-emerged as properties of the same vocal gesture.

Keywords

Language evolution, displaced reference, cooperative breeding, proper names, concept cells, vocal self-regulation, maternal separation, proto-syntax, neoteny, semiotics, Default Mode Network, rigid designation

1. Introduction

One of the most persistent puzzles in the study of language evolution is the origin of **displaced reference** — the ability to refer to entities not present in the immediate perceptual environment. Explaining how such a capacity could emerge from the affective and indexical vocal signaling of our primate ancestors remains a central theoretical challenge.

The difficulty is not merely functional but semiotic. In Peircean sign theory (Short, 2007), animal signals are predominantly **indexical**: they stand in a direct causal or spatial relationship with their objects and cease when the stimulus disappears. A symbol, by contrast, maintains its referential link through convention, independently of the object's perceptual availability. The transition from index to symbol therefore requires not just a new kind of signal, but a new mode of reference — one that can survive the absence of what it refers to.

Numerous models have attempted to bridge this gap — ecological recruitment (Bickerton, 2009), shared intentionality (Tomasello, 2008), the symbolic threshold (Deacon, 1997) — yet these approaches typically presuppose the capacity to refer to absent entities rather than explaining how it first arose. The more elementary question remains: **what kind of vocal act could first maintain a stable referent in the absence of its perceptual stimulus?**

The **Name-Leash hypothesis** argues that the first displaced referential unit was a proper name — an individualized vocalization addressed to an absent infant. In cooperative breeding contexts characteristic of early *Homo* (Hrdy, 2009; Burkart et al., 2025), a mother foraging at a distance from her cached infant faces a persistent sensory vacuum. The hypothesis proposes that under this pressure, a vocalization originally used for infant-directed contact was rhythmically repeated in solitude as a form of affective self-regulation. Through such repetition, it maintained the mental representation of the absent infant and came to function as a stable displaced referent.

The proper name is a strong candidate for the first displaced signal because it designates a specific individual across all contexts — what Kripke (1980) termed a **rigid designator** — and because the neural machinery for invariant identity representation already exists in the primate brain (Quiñero et al., 2005). The aim of this paper is to outline a coherent mechanism connecting cooperative breeding, mother–infant vocal interaction, physiological changes associated with upright walking, and neural representations of individual identity into a parsimonious pathway from affective vocal regulation to the first displaced symbolic reference.

2. Displacement: The Hockett–Bickerton Framework and Its Limits

Hockett (1960) defined displacement as the ability to communicate about things remote in space or time. Bickerton (2009) elevated this from one feature among many to the central problem of anthropogenesis, arguing that protolanguage required a decisive break from sensory-bound primate signaling, driven by the need to recruit coalition members to distant scavenging opportunities. (The protolanguage concept itself was introduced in Bickerton, 1990.)

Bickerton's contribution is indispensable: he correctly insists that any serious account of language origins must explain how communication escaped the "here and now." However, the recruitment-to-food scenario does not make this escape compelling. Pointing, pantomime, and leading — all perceptually grounded — would seem sufficient for guiding conspecifics to a nearby carcass. More fundamentally, there is no principled reason to privilege food acquisition over reproduction or offspring survival as the evolutionary bottleneck.

The semiotic objection is deeper still. Bickerton rejected the indexical route as a dead end and proposed that displacement arose through **iconic** signs — vocal imitations of absent objects. But this contains a logical circularity: to imitate the sound of an absent object, the imitator must already hold a mental representation of it — that is, displacement must already be in place. The icon presupposes the very capacity it is meant to explain.

What Bickerton did not consider is that the index itself might be *transformed* under specific pressures — not abandoned in favor of a new sign type, but reshaped from within.

3. The Semiotic Argument: From Broken Index to Proper Name

3.1 The Broken Index

Consider the simplest communicative situation available to a social animal: an indexical gesture directed at a present object — a contact call, a reaching gesture, a gaze toward a companion. In Peircean terms, these signs depend on a physical connection to a present referent (Short, 2007).

Now consider what happens when the referent disappears. The mother turns to look for her infant, but it has been taken by an allomother. The contact call meets no response. The indexical sign has lost its object.

This moment of rupture — the **broken index** — is not merely a communicative failure. It is a semiotic crisis that creates evolutionary pressure. The affective system demands resolution; something must fill the gap left by the absent referent.

3.2 The Proper Name as Repaired Index

The hypothesis proposes that this gap was filled by a vocal substitute: the rhythmic repetition of an individualized sound associated with the absent infant — a proto-name. This vocalization preserves the essential property of the index (it points to a specific individual) while acquiring

a new property: it functions in the individual's absence. It is a **portable pointing gesture** — an index detached from its perceptual anchor and converted into a symbol.

This analysis converges with Kripke's (1980) account of proper names as **rigid designators**: expressions whose reference remains fixed across all contexts. Crucially, a rigid designator carries no descriptive content requiring verification against the current environment: unlike "there is a leopard," which demands perceptual confirmation, a name activates its referent's representation independently of any state of affairs — making it the only signal type that can function precisely when there is nothing to perceive.

3.3 Why Not a Common Noun?

Several considerations make the proper name more plausible than a categorical label as the first displaced signal:

No abstraction required. A common noun classifies across instances; a proper name refers to a single individual already represented by a dedicated neural ensemble (a concept cell), making it cognitively the cheapest possible displaced signal.

No context verification. As a rigid designator, the name carries no information about the environment and requires no perceptual check. It simply holds an identity.

Affective subsidy. Separation anxiety provides the emotional energy that sustains the representation in working memory. What would otherwise be effortful attention is driven automatically by the limbic system.

Univocal selection. In the situation of separation, there is only one salient referent — the absent child. The mapping is one object — one label, the simplest possible referential structure.

No deception problem. Known cases of displacement-like behavior in animals — such as false alarm calls used by fork-tailed drongos (Flower, 2011) — are adversarial: the signal makes a falsifiable claim about the world. The proper name makes no such claim; there is nothing to falsify, no incentive for skepticism. The name-leash is a benign self-deception — a vocal act creating the illusion of presence to bridge a real absence — but unlike adversarial deception, this benefits both parties: it regulates the mother's anxiety, and when transmitted through the allomother, functions as an implicit promise of return.

4. Physiological Prerequisites: Upright Walking

The broken index could only be "repaired" by a proto-name if the hominin body was capable of sustaining rhythmic, voluntary vocal output independently of locomotion. Habitual upright walking (bipedalism) provided this capacity through three linked adaptations.

First, **decoupled respiration**. In quadrupeds, locomotion and breathing are mechanically coupled; each stride compresses the thorax and forces exhalation. Upright walking freed the thorax, permitting voluntary control of expiratory timing and duration (Provine, 2000) — the prerequisite for sustained rhythmic vocalization.

Second, **autonomic adaptation to orthostatic stress**. Maintaining cerebral perfusion against gravity required enhanced vagal tone and refined HRV regulation. This created an autonomic infrastructure co-optable for stress management, including vocal affect regulation. Empirical evidence links paced exhalation and rhythmic vocalization to increased parasympathetic influence (Bernardi et al., 2001; Vickhoff et al., 2013; Gerritsen & Band, 2018), consistent with Porges's (2011) polyvagal framework linking laryngeal motor control to autonomic regulation.

Third, **the exaptive platform**. Together, these changes — originally selected for biomechanical demands — constituted a platform for vocal self-regulation: a mother could now regulate her affect through rhythmic vocalization while foraging at a distance from her infant.

The paleoanthropological record is consistent with this timing. By ~2.0–1.8 million years ago, *H. ergaster/erectus* exhibits full terrestrial bipedalism and the earliest "home base" behavior — precisely the window when cooperative breeding would have generated the selective pressure for the name-leash mechanism.

5. Cooperative Breeding and the Affective Gap

5.1 The Cooperative Breeding Model

Hrady (2009) argued that early *Homo* was characterized by **cooperative breeding** — a strategy in which individuals other than the biological mother contribute substantially to infant care, freeing the mother for extended foraging. Burkart et al. (2025) strengthen this model by arguing that cooperative breeding was a likely **early catalyst** of the human evolutionary feedback loop, preceding and enabling subsequent increases in brain size.

5.2 The Affective Gap

Cooperative breeding creates a situation with no parallel in other great apes: daily physical separation between a mother and her infant. The critical feature is the **non-fungibility** of the infant — it is irreplaceable, making maternal separation anxiety qualitatively different from any ecological pressure. This non-fungibility is the emotional precondition for a signal that must be equally specific: a proper name, not a generic label.

Table 1

Comparison of the Name-Leash hypothesis and Bickerton's recruitment model

Feature	Bickerton's Recruitment Model	Name-Leash Hypothesis
Referent	Food source / Predator ("it")	Specific infant ("you")
Fungibility	Replaceable	Non-fungible
Frequency	Episodic, opportunistic	Daily, recurrent
Emotional valence	Moderate (hunger, fear)	Maximal (attachment)
Duration	Minutes	Hours (foraging bout)
Information content	High (location, type)	Minimal (identity only)
Deception structure	Adversarial (false claim)	Self-regulation (implicit promise)

5.3 Trust, Transfer, and the Proto-Credit Operation

The transfer of an infant to an allomother constitutes a **proto-credit operation**: the mother "lends" her non-fungible child, the allomother "holds" it in trust, the obligation of return is implicit. Cooperative breeding generates not merely separation anxiety but a **relational structure** (A entrusts X to B; B holds X for A; B returns X to A) that is, in embryonic form, the structure of social obligation.

6. The Core Mechanism: From Affective Mantra to Displaced Symbol

6.1 The Mantra Function

A mother in a cooperative breeding coalition leaves her infant with an allomother and departs to forage. The indexical connection is broken. To mitigate autonomic stress, she rhythmically repeats an individualized vocalization associated with her child — a proto-name embedded in motherese prosody.

This repetition performs a dual function. **Physiologically**, it constitutes a vocal-respiratory pattern that modulates autonomic state through the self-soothing mechanism described in Section 4. **Cognitively**, it activates and sustains identity-selective concept cells in the medial temporal lobe (Quiñan Quiroga et al., 2005), creating a **neurocognitive leash** — a sustained mental representation of the absent child.

6.2 The Birth of Displacement

The vocalization no longer reacts to a present stimulus; it **generates a mental state to compensate for a missing one**. In semiotic terms, the broken index has been repaired. In Kripke's terms, the proto-name has become a rigid designator: its reference is fixed to a specific individual independently of any state of affairs. The name does not describe where the child is — it simply **holds the identity**.

6.3 The Symbolon

The ancient Greek *symbolon* — a token broken in two, each half held by a different party as a sign of mutual recognition — captures the relational structure of the proto-name. The mother's vocalization is one half; the infant, held by the allomother, is the other. The name exists *for* reunion — it is a promissory sign whose existence presupposes the future act of return.

6.4 Conventionalization

The transition from private mantra to shared symbol occurs through mimesis. When the mother returns, the allomother — equipped with the primate capacity for vocal imitation — reproduces the sound. Accurate imitation has a calming effect, creating selective advantage for precise vocal matching. The proto-name becomes a **dyadically specific password** circulating among coalition members — a portable token of trust, decoupled from the referent but anchored through the emotional resonance of correct reproduction.

Figure 1 summarizes the proposed causal chain, from the two independent prerequisites through the semiotic crisis of the broken index to displaced reference and its syntactic elaboration.

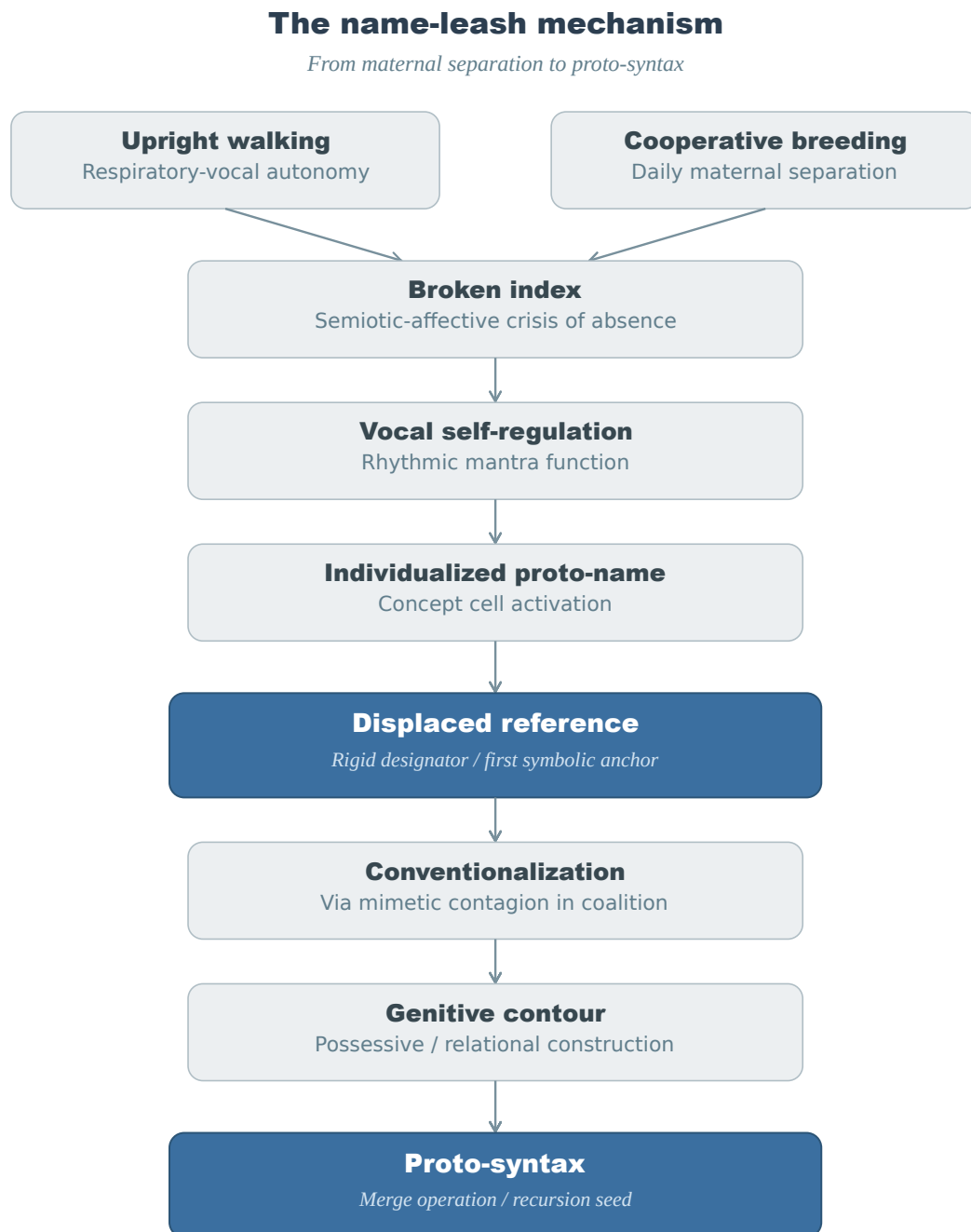


Figure 1. *The Name-Leash mechanism.* Two independent evolutionary prerequisites — respiratory-vocal autonomy from upright walking and daily maternal separation from cooperative breeding — converge to produce a semiotic crisis: the "broken index," in which the referent of an indexical signal disappears. Rhythmic vocal self-regulation repairs this break by sustaining the absent referent's neural representation via concept cell activation, producing the first displaced reference. Conventionalization through mimetic contagion transforms the private mantra into a shared symbol, whose inherent relational (genitive) structure provides the seed for proto-syntactic Merge.

7. Developmental Evidence

If the Name-Leash scenario captures a genuine cognitive-affective dynamic, its traces should appear in ontogeny. Three independent bodies of evidence converge on a single pattern.

7.1 The Babbling Peak

Delack's (1976) longitudinal data, as compiled in Locke (1993), show that between 33 and 43 weeks — the peak window for separation anxiety (Bowlby, 1969; Schaffer & Emerson, 1964) and the emergence of object permanence (Piaget, 1954) — solitary vocalizations peak at approximately 67% of all infant vocalizations, while vocalizations in the mother's presence drop to a minimum of approximately 20% (Figure 2). The inverse relationship is striking: at the developmental moment when separation anxiety is most intense, the infant vocalizes most when alone and least when the attachment figure is present. This temporal alignment suggests a functional role for babbling in infant self-regulation: the vocal-respiratory cycles of canonical babbling serve as an emergent self-soothing mechanism bridging the affective gap of the caregiver's absence (Stark, 1986).

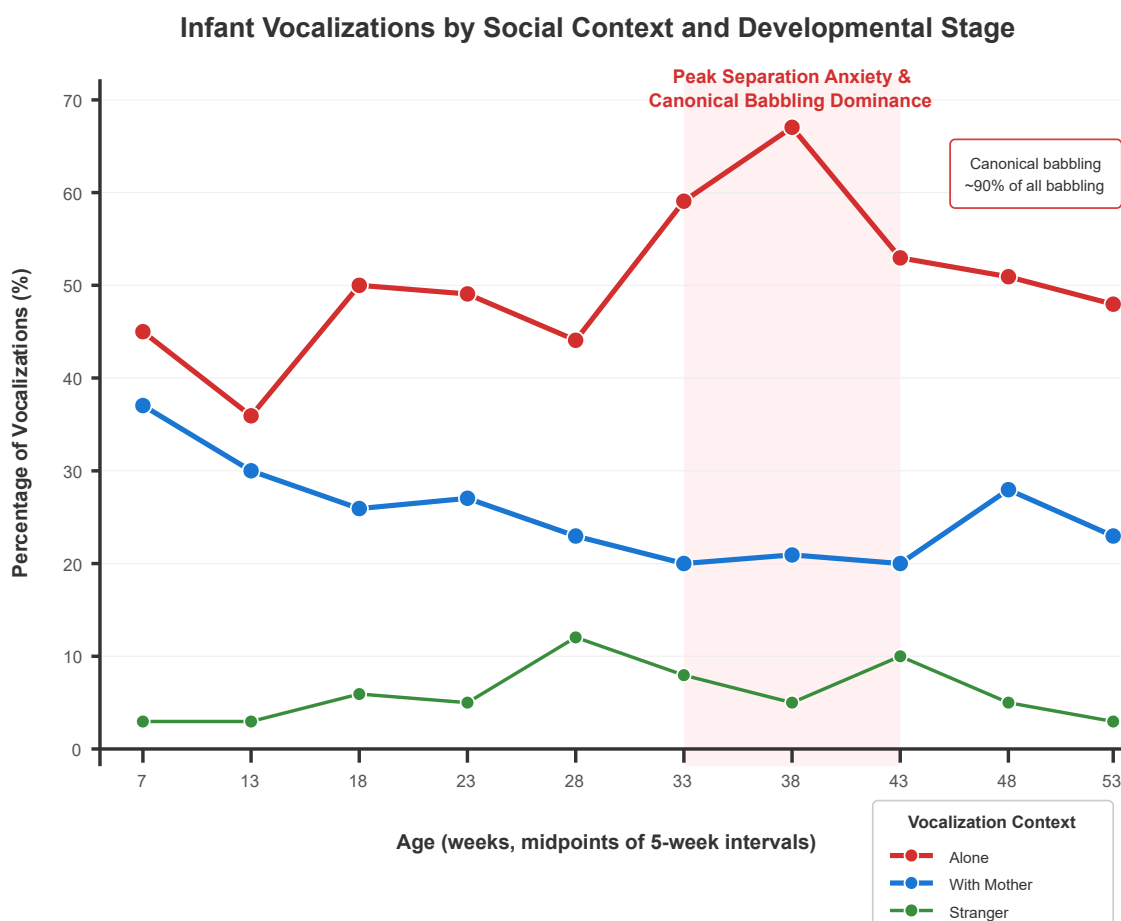


Figure 2. *Infant vocalizations by social context across the first year of life.* Data show the percentage of infant vocalizations occurring when alone (red), with mother (blue), or with a stranger (green) at successive developmental stages from 7 to 53 weeks. Data from Delack (1976), as compiled in Locke (1993, Fig. 4.1). The shaded zone (33–43 weeks) highlights the critical window of separation anxiety and canonical babbling dominance .

7.2 Freud's Fort/Da

In *Beyond the Pleasure Principle* (1920), Freud describes a child repeatedly throwing and retrieving a spool, uttering "Fort!" ("gone!") and "Da!" ("there!") — symbolic mastery of the mother's departure and return. The minimal symbolic act arises from the experience of absence, and its function is not referential but **regulatory**: it manages the affect generated by separation. The game's structure — presence → absence → return — is the minimal unit of diachrony.

7.3 Porshnev's "First Word"

Porshnev (1974) identifies three features of the child's first articulated word (11–13 months): it is **singular** (preceding any system of oppositions), **functionally regulatory** (the child imitates a sound complex associated with the inhibition of its own actions, not a referential label), and the prototypical situation involves **denial of maternal contact** — refusal of the breast, refusal to be picked up. "Mama," when it appears as the first word, does not name the mother; it reproduces the vocal pattern of the most emotionally charged prohibition. This is the ontogenetic analogue of the separation stress the hypothesis posits as the phylogenetic driver.

7.4 Convergence

All three lines show the same pattern: the earliest proto-symbolic vocal acts arise from maternal separation, serve affective self-regulation rather than information transfer, and are rhythmic, repetitive, and self-directed — precisely the dynamic the Name-Leash hypothesis predicts.

8. Comparative Evidence: Vocal Individuation and Vocal Mimesis

The name-leash mechanism requires two biological capacities rarely found together: **vocal individuation** (individualized signatures) and **vocal mimesis** (reproducing another's vocalization).

Bottlenose dolphins produce individualized signature whistles and modify them in motherese style for their calves (Sayigh et al., 2023); sac-winged bats similarly produce pup-directed vocalizations differing in timbre (Fernandez & Knörnschild, 2020). Both exhibit vocal individuation but lack the sophisticated social mimesis of primates — their "names" remain functional beacons, not symbolic tools circulating among third parties.

Chimpanzees exhibit powerful **vocal convergence**: males adjust pant-hoots to match frequent associates, creating group-level acoustic signatures (Mitani & Brandt, 1994; Lombrey & Fröhlich, 2025). This is synchronic mimesis — group unification in the "here and now" — but it targets the group, not the individual. Without cooperative breeding, this mimetic capacity never turns to the task of holding a specific absent individual in mind.

Common marmosets (*Callithrix jacchus*) — cooperative breeders — provide a critical bridge: contingent vocal feedback from caregivers bootstraps vocal development in infants (Takahashi et al., 2017). Specifically the *individualized, responsive* interaction drives learning — supporting the claim that cooperative breeding contexts favor directed vocal communication. Table 2 summarizes the comparative landscape:

Table 2*Distribution of prerequisites for displaced reference across species*

Feature	Dolphins/Bats	Chimpanzees	Marmosets	Early <i>Homo</i>
Vocal individuation	YES	NO	YES	YES
Vocal mimesis	LIMITED	YES (group)	LIMITED	YES
Cooperative breeding	NO/partial	NO	YES	YES
Displaced reference	NO	NO	NO	YES

Displaced reference emerges only where all three conditions co-occur.

9. Neural Substrate: Concept Cells and the Left Temporal Pole

9.1 Concept Cells

Concept cells — sparse, highly selective neurons in the medial temporal lobe (Quiñero et al., 2005, 2012, 2017) — fire for a specific known individual regardless of sensory modality. They provide an ultra-compact, modality-independent identity code — precisely what a rigid designator requires. Similar identity-selective neurons exist in macaques (Chang & Tsao, 2017; Freiwald & Tsao, 2010), indicating that the primate brain was pre-equipped for individualized representation. The direct connection between concept cells and the amygdala ensures that identity representations of attachment figures are **affectively loaded** — the neural correlate of the separation anxiety driving the name-leash mechanism.

9.2 The Left Temporal Pole

Neuropsychological evidence demonstrates that proper names and common nouns are processed by partially distinct systems (Semenza, 2006; Damasio et al., 1996). Focal damage to the **left temporal pole (LTP)** produces selective anomia for proper names while sparing semantic knowledge and common nouns — indicating that proper names are a **cognitively distinct referential class**, not a subcategory of nouns. The LTP is also a node of the **Default Mode Network** (Buckner et al., 2008) — the network most active during social cognition and mental simulation of absent others — placing the name-retrieval system at the hub of the brain's "displacement network."

9.3 Evolutionary Mismatch

Names of emotionally close individuals are almost never forgotten in healthy adults, consolidated through limbic neurochemistry. The system's apparent "fragility" — frequent tip-of-the-tongue failures — manifests only at its periphery, when modern humans attempt to store hundreds of names for which the system was never designed. This mismatch is evidence of the system's original specialization for a small number of affectively significant attachment figures.

10. The Genitive Contour as Proto-Syntactic Seed

The proto-name was born as an inherently relational unit — encoding not merely identity but **inalienable possession**. The infant *was* "Mother's [Name]" before "[Name]" existed as an independent label. Weiner (1992) described inalienable possessions in Oceanic societies — objects that remain defined by their relation to their original owner. The infant in cooperative breeding is the paradigm case: temporarily held by an allomother, but *remaining* the mother's child. The situation "my child is not with me" is, in embryonic form, the genitive construction.

This relational character manifests prosodically in motherese: the name is embedded in contours of possession, search, or distress, much as patronymic surnames (Ivan-ov, Peter-son) fossilize genitive morphology. As the proto-name becomes conventionalized, it acquires the potential for **decomposition**: "[Mother's-child]" → "[Mother]" + "[child]" → recombination → the first combinatorial structure. Merge, in its minimal sense, arises as a byproduct of the decomposition of the genitive structure.

Recursive nesting of social relations ("Mother of [Name]," "Sister of [Mother of [Name]]") becomes tractable only with post-reproductive longevity and grandmothering (Caspari & Lee, 2004). Recursion is not an immediate consequence of the name-leash but a later exploitation of its relational architecture — the genitive contour is the seed; recursive kinship terminology is the tree.

11. Integration Within Broader Anthropogenesis

The name-leash mechanism has implications beyond the first displaced signal. The rhythmic vocalization encodes the first **social obligation** — a deferred duty that cannot be immediately reciprocated because the non-fungible child is not exchangeable. The mother's vocalization is not only a coping mechanism but an implicit promise to return — a self-binding act. That the first symbolic act may have been simultaneously referential (naming) and promissory (committing to return) suggests that **reference and obligation co-emerged as properties of the same vocal gesture**.

Non-fungibility creates debt that cannot be discharged instantly; debt generates **experienced duration** — time as structured expectation rather than perception of external cycles. The expansion of alloparenting under neotenization (Hare, 2017; Somel et al., 2009) may have generalized the mother–infant dynamic to broader social relationships, making displaced reference and promissory commitment available to all group members. This mechanism supplies the missing evolutionary scenario for Deacon's (1997) symbolic threshold and the first argument slot for Merge. The full development of this argument will be presented in a forthcoming monograph.

12. Comparison with Existing Models

Bickerton (2009; see also 1990) correctly identified displacement as the central problem but located pressure in ecological recruitment; as argued above, this does not compellingly select for offline representation, and the iconic path is logically circular.

Falk (2004, 2009) proposed motherese as an "audio leash" for contact maintenance. The Name-Leash hypothesis shifts the critical moment from contact (Falk's leash is intact) to separation (the leash is broken), and relies on infant transfer to allomothers rather than the questionable practice of laying infants on the ground.

Tomasello (2008) emphasizes shared intentionality — a synchronic cooperative stance that does not, by itself, explain how reference operates in the referent's absence.

Deacon (1997) offered a sophisticated Peircean account of symbolic reference but no specific evolutionary scenario for the first displaced symbol. The broken index repaired by the proper name supplies this missing scenario.

Limitations

The hypothesis is conceptual: it proposes a plausible mechanism connecting cooperative breeding, maternal separation anxiety, and displaced symbolic reference, but does not claim direct empirical demonstration.

The causal chain is inferred from convergent evidence across disciplines rather than from any single line of direct proof; no archaeological evidence can attest to Pleistocene vocalizations. The hypothesis depends on the chronological priority of cooperative breeding, which, while strongly supported (Hrdy, 2009; Burkart et al., 2025), remains debated. The developmental parallels (Section 7) are suggestive but do not constitute proof of phylogenetic recapitulation. The semiotic argument (Section 3) identifies a possible pathway from index to symbol, not the only one.

Despite these limitations, the hypothesis generates expectations evaluable against future comparative, developmental, and neuroimaging data: Do other cooperative breeders show elevated individualized vocalizations during separation? Do concept cells show heightened activation during vocal self-regulation in the absence of the represented individual? Do infant babbling corpora contain genitive-like prosodic contours? These are directions for inquiry rather than predictions in the strong sense, but they indicate the empirical space the hypothesis opens.

Competing Interests

The author declares no competing interests.

Data Availability

This paper is a theoretical contribution and does not report original empirical data. All referenced data are available in the cited publications. Figure 2 is based on data from Delack (1976) as compiled in Locke (1993, Fig. 4.1).

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